

Fig. 4.1.2. P does not separate the region O of $\mathbb{R}^2 \setminus (X_1 \cup X_2)$

It remains to introduce a few terms and facts that will be used only once, when we consider notions of equivalence for graph drawings in Chapter 4.3.

As usual, we denote by S^n the n -dimensional sphere, the set of points in $\mathbb{R}^{n+1}$ at distance 1 from the origin. The 2-sphere minus its ‘north pole’ $(0, 0, 1)$ is homeomorphic to the plane; let us choose a fixed such homeomorphism $\pi: S^2 \setminus \{(0, 0, 1)\} \rightarrow \mathbb{R}^2$ (for example, stereographic projection). If $P \subseteq \mathbb{R}^2$ is a polygon and O is the bounded region of $\mathbb{R}^2 \setminus P$, let us call $C := \pi^{-1}(P)$ a *circle on S^2* , and the sets $\pi^{-1}(O)$ and $S^2 \setminus \pi^{-1}(P \cup O)$ the *regions* of $S^2 \setminus C$.

Our last tool is the theorem of Jordan and Schoenflies, again adapted slightly for our purposes:

Theorem 4.1.4. *Let $\varphi: C_1 \rightarrow C_2$ be a homeomorphism between two circles on S^2 , let O_1 be a region of $S^2 \setminus C_1$, and let O_2 be a region of $S^2 \setminus C_2$. Then φ can be extended to a homeomorphism $C_1 \cup O_1 \rightarrow C_2 \cup O_2$.*

4.2 Plane graphs

A *plane graph* is a pair (V, E) of finite sets with the following properties (the elements of V are again called *vertices*, those of E *edges*):

- (i) $V \subseteq \mathbb{R}^2$;
- (ii) every edge is an arc between two vertices;
- (iii) different edges have different sets of endpoints;
- (iv) the interior of an edge contains no vertex and no point of any other edge.

A plane graph (V, E) defines a graph G on V in a natural way. As long as no confusion can arise, we shall use the name G of this abstract graph also for the plane graph (V, E) , or for the point set $V \cup \bigcup E$; similar notational conventions will be used for abstract versus plane edges, for subgraphs, and so on.¹

¹ However, we shall continue to use $\setminus$ for differences of point sets and $-$ for graph differences – which may help a little to keep the two apart.

When G is a plane graph, we call the regions of $\mathbb{R}^2 \setminus G$ the *faces* of G . These are open subsets of $\mathbb{R}^2$ and hence have their frontiers in G . Since G is bounded – i.e., lies inside some sufficiently large disc D – exactly one of its faces is unbounded, the face that contains $\mathbb{R}^2 \setminus D$. This face is the *outer face* of G ; the other faces are its *inner faces*. We denote the set of faces of G by $F(G)$.

The faces of plane graphs and their subgraphs are related in the obvious way:

Lemma 4.2.1. *Let G be a plane graph, $f \in F(G)$ a face, and $H \subseteq G$ a subgraph.*

[4.4.3]

- (i) H has a face f' containing f .
- (ii) If the frontier of f lies in H , then $f' = f$.

Proof. (i) Clearly, the points in f are equivalent also in $\mathbb{R}^2 \setminus H$; let f' be the equivalence class of $\mathbb{R}^2 \setminus H$ containing them.

(ii) Recall from Section 4.1 that any arc between f and $f' \setminus f$ meets the frontier X of f . If $f' \setminus f \neq \emptyset$ then there is such an arc inside f' , whose points in X do not lie in H . Hence $X \not\subseteq H$. $\square$

In order to lay the foundations for the (easy but) rigorous introduction to plane graphs that this section aims to provide, let us descend once now into the realm of truly elementary topology of the plane, and prove what seems entirely obvious:² that the frontier of a face of a plane graph G is always a subgraph of G – not, say, half an edge.

The following lemma states this formally, together with two similarly ‘obvious’ properties of plane graphs:

Lemma 4.2.2. *Let G be a plane graph and e an edge of G .*

[4.5.1]
[4.5.2]
[12.7.4]

- (i) If X is the frontier of a face of G , then either $e \subseteq X$ or $X \cap \dot{e} = \emptyset$.
- (ii) If e lies on a cycle $C \subseteq G$, then e lies on the frontier of exactly two faces of G , and these are contained in distinct faces of C .
- (iii) If e lies on no cycle, then e lies on the frontier of exactly one face of G .

Proof. We prove all three assertions together. Let us start by considering one point $x_0 \in \dot{e}$. We show that x_0 lies on the frontier of either exactly two faces or exactly one, according as e lies on a cycle in G or not. We then show that every other point in $\dot{e}$ lies on the frontier of exactly the same faces as x_0 . Then the endpoints of e will also lie on the frontier of

(4.1.1)
(4.1.3)

² Note that even the best intuition can only ever be ‘accurate’, i.e., coincide with what the technical definitions imply, inasmuch as those definitions do indeed formalize what is intuitively intended. Given the complexity of definitions in elementary topology, this can hardly be taken for granted.

these faces – simply because every neighbourhood of an endpoint of e is also the neighbourhood of an inner point of e .

Since $G \setminus \dot{e}$ is compact, we can find around every point $x \in \dot{e}$ an open disc D_x that meets G only in those (one or two) straight line segments that contain x .

Let us pick an inner point x_0 from a straight line segment $S \subseteq e$. Then $D_{x_0} \cap G = D_{x_0} \cap S$, so $D_{x_0} \setminus G$ is the union of two open half-discs. Since these half-discs do not meet G , they each lie in a face of G . Let us denote these faces by f_1 and f_2 ; they are the only faces of G with x_0 on their frontier, and they may coincide (Fig. 4.2.1).

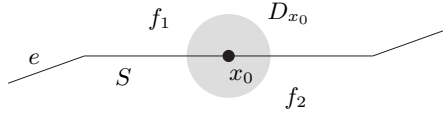


Fig. 4.2.1. Faces f_1, f_2 of G in the proof of Lemma 4.2.2

If e lies on a cycle $C \subseteq G$, then D_{x_0} meets both faces of C (Theorem 4.1.1). Since f_1 and f_2 are contained in faces of C by Lemma 4.2.1, this implies $f_1 \neq f_2$. If e does not lie on any cycle, then e is a bridge and thus links two disjoint point sets X_1, X_2 as in Lemma 4.1.3, with $X_1 \cup X_2 = G \setminus \dot{e}$. Clearly, $f_1 \cup \dot{e} \cup f_2$ is the subset of a face f of $G - e$. By Lemma 4.1.3, $f \setminus \dot{e}$ is a face of G , while $f_1, f_2 \subseteq f \setminus \dot{e}$ by definition of f . Since f_1 and f_2 are also faces of G , this implies $f_1 = f \setminus \dot{e} = f_2$.

Now consider any other point $x_1 \in \dot{e}$. Let P be the arc from x_0 to x_1 contained in e . Since P is compact, finitely many of the discs D_x with $x \in P$ cover P . Let us enumerate these discs as $D_0, \dots, D_n$ in the natural order of their centres along P ; adding D_{x_0} or D_{x_1} as necessary, we may assume that $D_0 = D_{x_0}$ and $D_n = D_{x_1}$. By induction on n , one easily proves that every point $y \in D_n \setminus e$ can be linked by an arc inside $(D_0 \cup \dots \cup D_n) \setminus e$ to a point $z \in D_0 \setminus e$ (Fig. 4.2.2); then y and z are equivalent in $\mathbb{R}^2 \setminus G$. Hence, every point of $D_n \setminus e$ lies in f_1 or in f_2 , so x_1 cannot lie on the frontier of any other face of G . Since both half-discs of $D_0 \setminus e$ can be linked to $D_n \setminus e$ in this way (swap the roles of D_0 and D_n), we find that x_1 lies on the frontier of both f_1 and f_2 . $\square$

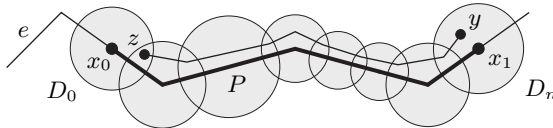


Fig. 4.2.2. An arc from y to D_0 , close to P

Corollary 4.2.3. *The frontier of a face is always the point set of a subgraph.* $\square$

The subgraph of G whose point set is the frontier of a face f is said to *bound* f and is called its *boundary*; we denote it by $G[f]$. A face is said to be *incident* with the vertices and edges of its boundary. By Lemma 4.2.1 (ii), every face of G is also a face of its boundary; we shall use this fact frequently in the proofs to come.

boundary
 $G[f]$

Proposition 4.2.4. *A plane forest has exactly one face.*

[4.6.1]

Proof. Use induction on the number of edges and Lemma 4.1.3. $\square$

(4.1.3)

With just one exception, different faces of a plane graph have different boundaries:

Lemma 4.2.5. *If a plane graph has different faces with the same boundary, then the graph is a cycle.*

[4.3.1]

Proof. Let G be a plane graph, and let $H \subseteq G$ be the boundary of distinct faces f_1, f_2 of G . Since f_1 and f_2 are also faces of H , Proposition 4.2.4 implies that H contains a cycle C . By Lemma 4.2.2 (ii), f_1 and f_2 are contained in different faces of C . Since f_1 and f_2 both have all of H as boundary, this implies that $H = C$: any further vertex or edge of H would lie in one of the faces of C and hence not on the boundary of the other. Thus, f_1 and f_2 are distinct faces of C . As C has only two faces, it follows that $f_1 \cup C \cup f_2 = \mathbb{R}^2$ and hence $G = C$. $\square$

(4.1.1)

Proposition 4.2.6. *In a 2-connected plane graph, every face is bounded by a cycle.*

[4.3.1]
[4.4.3]
[4.5.1]
[4.5.2]

Proof. Let f be a face in a 2-connected plane graph G . We show by induction on $\|G\|$ that $G[f]$ is a cycle. If G is itself a cycle, this holds by Theorem 4.1.1; we therefore assume that G is not a cycle.

(3.1.1)
(4.1.1)
(4.1.2)

By Proposition 3.1.1, there exist a 2-connected plane graph $H \subseteq G$ and a plane H -path P such that $G = H \cup P$. The interior of P lies in a face f' of H , which by the induction hypothesis is bounded by a cycle C .

H
 P
 f', C

If $G[f] \subseteq H$, then f is also a face of H (Lemma 4.2.1 (ii)), and we are home by the induction hypothesis. If $G[f] \not\subseteq H$, then $G[f]$ meets $P \setminus H$, so $f \subseteq f'$ and $G[f] \subseteq C \cup P$ (why?). By Lemma 4.2.1 (ii), then, f is a face of $C \cup P$ and hence bounded by a cycle (Lemma 4.1.2 (i)). $\square$

In a 3-connected graph, we can identify the face boundaries among the other cycles in purely combinatorial terms:

Proposition 4.2.7. *The face boundaries in a 3-connected plane graph are precisely its non-separating induced cycles.*

[4.3.2]
[4.5.2]

(3.3.6) *Proof.* Let G be a 3-connected plane graph, and let $C \subseteq G$. If C is a non-separating induced cycle, then by the Jordan curve theorem its two faces cannot both contain points of $G \setminus C$. Therefore it bounds a face of G .

(4.1.1) C, f Conversely, suppose that C bounds a face f . By Proposition 4.2.6, C is a cycle. If C has a chord $e = xy$, then the components of $C - \{x, y\}$ are linked by a C -path in G , because G is 3-connected. This path and e both run through the other face of C (not f) but do not intersect, a contradiction to Lemma 4.1.2 (ii).

It remains to show that C does not separate any two vertices $x, y \in G - C$. By Menger's theorem (3.3.6), x and y are linked in G by three independent paths. By Lemma 4.2.1 (i), f lies inside a face of their union, and by Lemma 4.1.2 (i) this face is bounded by only two of the paths. The third therefore avoids f and its boundary C . $\square$

maximal
plane graph

A plane graph G is called *maximally plane*, or just *maximal*, if we cannot add a new edge to form a plane graph $G' \supsetneq G$ with $V(G') = V(G)$.

plane
triangulation

We call G a *plane triangulation* if every face of G (including the outer face) is bounded by a triangle.

[4.4.1]
[5.4.2] **Proposition 4.2.8.** *A plane graph of order at least 3 is maximally plane if and only if it is a plane triangulation.*

(4.1.2) *Proof.* Let G be a plane graph of order at least 3. It is easy to see that if every face of G is bounded by a triangle, then G is maximally plane. Indeed, any additional edge e would have its interior inside a face of G and its ends on the boundary of that face. Hence these ends are already adjacent in G , so $G \cup e$ cannot satisfy condition (iii) in the definition of a plane graph.

f Conversely, assume that G is maximally plane and let $f \in F(G)$ be a face; let us write $H := G[f]$. Since G is maximal as a plane graph, $G[H]$ is complete: any two vertices of H that are not already adjacent in G could be linked by an arc through f , extending G to a larger plane graph. Thus $G[H] = K^n$ for some n – but we do not know yet which edges of $G[H]$ lie in H .

H Let us show first that H contains a cycle. If not, then $G \setminus H \neq \emptyset$: by $G \supseteq K^n$ if $n \geq 3$, or else by $|G| \geq 3$. On the other hand we have $f \cup H = \mathbb{R}^2$ by Proposition 4.2.4 and hence $G = H$, a contradiction.

n Since H contains a cycle, it suffices to show that $n \leq 3$: then $H = K^3$ as claimed. Suppose $n \geq 4$, and let $C = v_1 v_2 v_3 v_4 v_1$ be a cycle in $G[H]$ ($= K^n$). By $C \subseteq G$, our face f is contained in a face f_C of C ; let f'_C be the other face of C . Since the vertices v_1 and v_3 lie on the boundary of f , they can be linked by an arc whose interior lies in f_C and avoids G . Hence by Lemma 4.1.2 (ii), the plane edge $v_2 v_4$ of $G[H]$ runs through f'_C rather than f_C (Fig. 4.2.3). Analogously, since $v_2, v_4 \in G[f]$, the edge $v_1 v_3$ runs through f'_C . But the edges $v_1 v_3$ and $v_2 v_4$ are disjoint, so this contradicts Lemma 4.1.2 (ii). $\square$

C, v_i

f_C, f'_C

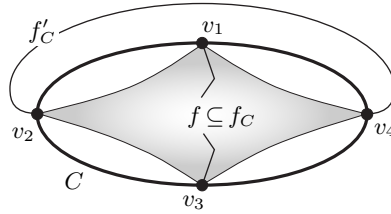


Fig. 4.2.3. The edge v_2v_4 of G runs through the face f'_C

The following classic result of Euler (1752) – here stated in its simplest form, for the plane – marks one of the common origins of graph theory and topology. The theorem relates the number of vertices, edges and faces in a plane graph: taken with the correct signs, these numbers always add up to 2. The general form of Euler's theorem asserts the same for graphs suitably embedded in other surfaces, too: the sum obtained is always a fixed number depending only on the surface, not on the graph, and this number differs for distinct (orientable closed) surfaces. Hence, any two such surfaces can be distinguished by a simple arithmetic invariant of the graphs embedded in them!³

Let us then prove Euler's theorem in its simplest form:

Theorem 4.2.9. (Euler's Formula)

Let G be a connected plane graph with n vertices, m edges, and ℓ faces. Then

$$n - m + \ell = 2.$$

Proof. We fix n and apply induction on m . For $m \leq n - 1$, G is a tree and $m = n - 1$ (why?), so the assertion follows from Proposition 4.2.4. (1.5.1)

Now let $m \geq n$. Then G has an edge e that lies on a cycle; let $G' := G - e$. By Lemma 4.2.2 (ii), e lies on the boundary of exactly two faces f_1, f_2 of G , and as the points in $\dot{e}$ are all equivalent in $\mathbb{R}^2 \setminus G'$, there is a face f_e of G' containing $\dot{e}$. We show that (1.5.2)

$$F(G) \setminus \{f_1, f_2\} = F(G') \setminus \{f_e\}; \quad (*)$$

then G' has exactly one face and one edge less than G , and so the assertion follows from the induction hypothesis for G' .

For a proof of $(*)$ let first $f \in F(G) \setminus \{f_1, f_2\}$ be given. By Lemma 4.2.2 (i) we have $G[f] \subseteq G \setminus \dot{e} = G'$, and hence $f \in F(G')$ by Lemma 4.2.1 (ii). As clearly $f \neq f_e$, this establishes the forward inclusion in $(*)$.

Conversely, consider any face $f' \in F(G') \setminus \{f_e\}$. Clearly $f' \neq f_1, f_2$, and $f' \cap \dot{e} = \emptyset$. Hence every two points of f' lie in $\mathbb{R}^2 \setminus G$ and are

³ This fundamental connection between graphs and surfaces lies at the heart of the proof of the famous Robertson-Seymour *graph minor theorem*; see Chapter 12.7.

equivalent there, so G has a face f containing f' . By Lemma 4.2.1 (i), however, f lies inside a face f'' of G' . Thus $f' \subseteq f \subseteq f''$ and hence $f' = f = f''$, since both f' and f'' are faces of G' . $\square$

[4.4.1]
[5.1.2]
[7.3.5]

Corollary 4.2.10. *A plane graph with $n \geq 3$ vertices has at most $3n - 6$ edges. Every plane triangulation with n vertices has $3n - 6$ edges.*

Proof. By Proposition 4.2.8 it suffices to prove the second assertion. In a plane triangulation G , every face boundary contains exactly three edges, and every edge lies on the boundary of exactly two faces (Lemma 4.2.2). The bipartite graph on $E(G) \cup F(G)$ with edge set $\{ef \mid e \subseteq G[f]\}$ thus has exactly $2|E(G)| = 3|F(G)|$ edges. According to this identity we may replace ℓ with $2m/3$ in Euler's formula, and obtain $m = 3n - 6$. $\square$

Euler's formula can be useful for showing that certain graphs cannot occur as plane graphs. The graph K^5 , for example, has $10 > 3 \cdot 5 - 6$ edges, more than allowed by Corollary 4.2.10. Similarly, $K_{3,3}$ cannot be a plane graph. For since $K_{3,3}$ is 2-connected but contains no triangle, every face of a plane $K_{3,3}$ would be bounded by a cycle of length ≥ 4 (Proposition 4.2.6). As in the proof of Corollary 4.2.10 this implies $2m \geq 4\ell$, which yields $m \leq 2n - 4$ when substituted in Euler's formula. But $K_{3,3}$ has $9 > 2 \cdot 6 - 4$ edges.

Clearly, along with K^5 and $K_{3,3}$ themselves, their subdivisions cannot occur as plane graphs either:

[4.4.5]
[4.4.6]
[5.1.2]

Corollary 4.2.11. *A plane graph contains neither K^5 nor $K_{3,3}$ as a topological minor.* $\square$

Surprisingly, it turns out that this simple property of plane graphs identifies them among all other graphs: as Section 4.4 will show, an arbitrary graph can be drawn in the plane if and only if it has no (topological) K^5 or $K_{3,3}$ minor.

4.3 Drawings

planar
embedding

drawing

An embedding in the plane, or *planar embedding*, of an (abstract) graph G is an isomorphism between G and a plane graph H . The latter will be called a *drawing* of G . We shall not always distinguish notationally between the vertices and edges of G and of H . In this section we investigate how two planar embeddings of a graph can differ.

How should we measure the likeness of two embeddings $\varrho: G \rightarrow H$ and $\varrho': G \rightarrow H'$ of a planar graph G ? An obvious way to do this is to consider the canonical isomorphism $\sigma := \varrho' \circ \varrho^{-1}$ between H and H' as abstract graphs, and ask how much of their position in the plane